PHYS121 Integrated Science-Physics

W6T1 the Kinetic Theory of Gases

References:
[1] David Halliday, Jearl Walker, Resnick Jearl, ‘Fundamentals of Physics’, (Wiley, 2018)
[2] Doug Giancoli, ‘Physics for Scientists and Engineers with modern physics’, (Pearson, 2009)
[3] Hugh D. Young, Roger A. Freedman, ‘University Physics with Modern Physics’, (Pearson, 2012)
And others specified when needed.

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18.6.4. Which one of the following statements is the best explanation for the fact that metal pipes that carry water often burst during cold winter months?

a) Both the metal and the water expand, but the water expands to a greater extent.

b) Water contracts upon freezing while the metal expands at lower temperatures.

c) The metal contracts to a greater extent than the water.

d) The interior of the pipe contracts less than the outside of the pipe.

e) Water expands upon freezing while the metal contracts at lower temperatures.

Learning Outcomes

- Apply the ideal gas law to solve problems
- Explain the microscopic view of atom/molecule movement.
- Explain vapor pressure and humidity
- Describe Maxwell's distribution and find the rms speed and mean free path.
- Describe phase transition processes and mole specific heat.
- Explain the adiabatic process and the Van der Waals equation of state.

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Avogadro's Number

The **kinetic theory of gases** relates the macroscopic properties of gases to the microscopic properties of gas molecules.

One **mole** of a substance contains N_A (**Avogadro's number**) elementary units (usually atoms or molecules), where N_A is found experimentally to be $N_A = 6.02 \times 10^{23}\,\text{mol}^{-1}$

The mass per mole M of a substance is related to the mass m of an individual molecule of the substance by $M = mN_\text{A}$

The number of moles n contained in a sample of mass M_sam, consisting of N molecules, is related to the molar mass M of the molecules and to Avogadro's number N_A as given by

$$n = \frac{M_\text{sam}}{M} = \frac{M_\text{sam}}{mN_\text{A}}.$$

Ideal Gases

An ideal gas is one for which the pressure *p*, volume *V*, and temperature *T* are related by

$$pV = nRT$$

Here *n* is the number of moles of the gas present and R is a constant (8.31 J/mol·K) called the gas constant.

The Second Expression for the law is:

$$pV = NkT$$

where *k* is the Boltzmann constant

(*a*)

(*b*)

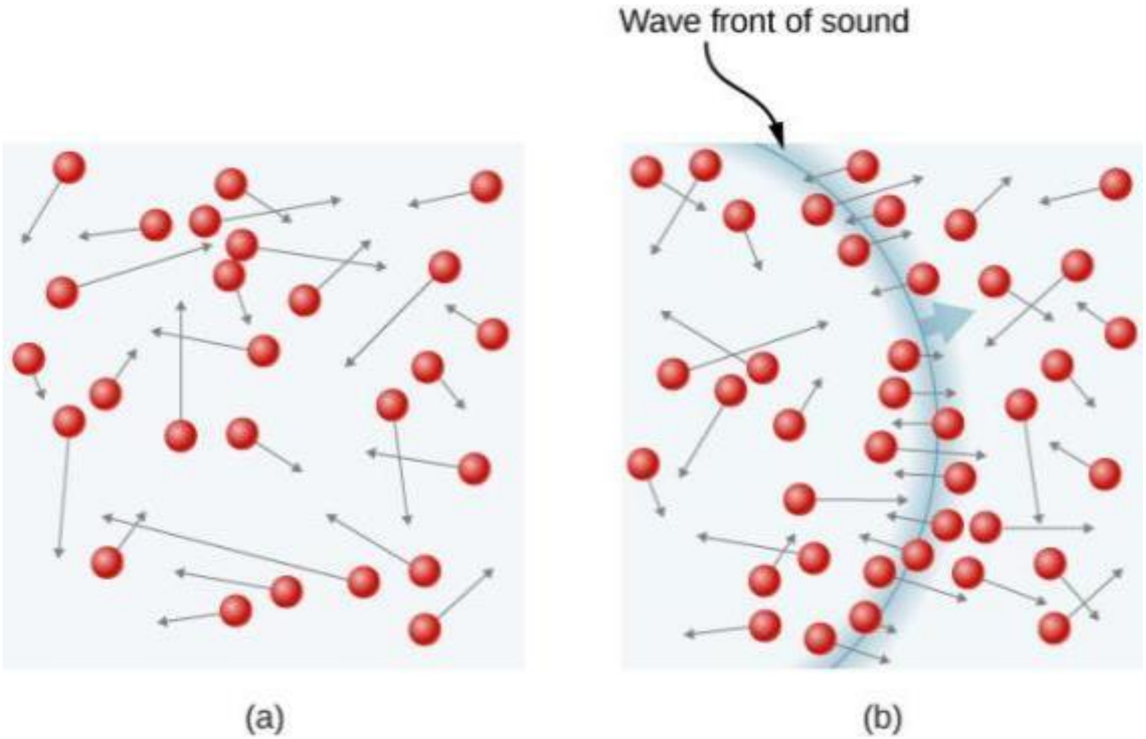
Courtesy www.doctorslime.com

Large steel tank crushed by atmospheric pressure. A dramatic example of ideal gas law

Example: Helium balloon.

A helium party balloon, assumed to be a perfect sphere, has a radius of 18.0 cm. At room temperature (20 ^{o}C), its internal pressure is 1.05 atm. Find the number of moles of helium in the balloon and the mass of helium needed to inflate the balloon to these values.

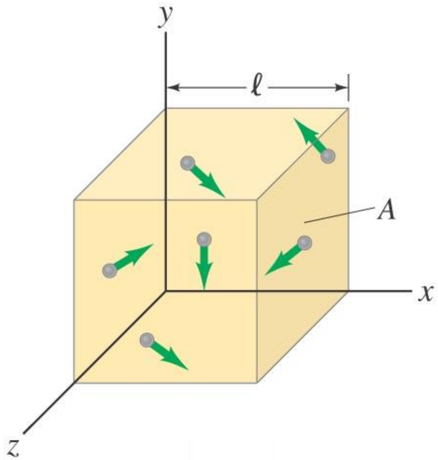
FIGURE 2.11



(a) In an ordinary gas, so many molecules move so fast that they collide billions of times every second.

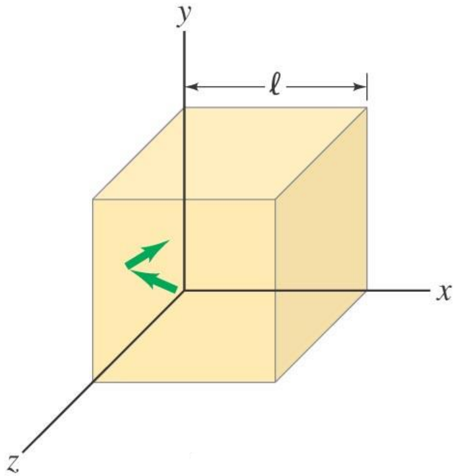
(b) Individual molecules do not move very far in a small amount of time, but disturbances like sound waves are transmitted at speeds related to the molecular speeds.

the Molecular Interpretation of Temperature



The **force** exerted on the wall by the collision of one molecule is

$$F = \frac{\Delta(mv)}{\Delta t} = \frac{2mv_x}{2\ell/v_x} = \frac{mv_x^2}{\ell}.$$



Then the force due to **all** molecules colliding with that wall is

$$F = \frac{m}{\ell} N\overline{v_x^2}.$$

The averages of the squares of the speeds in all three directions are equal:

$$F = \frac{m}{\ell} N \frac{\overline{v^2}}{3}.$$

So the pressure is:

$$P = \frac{1}{3} \frac{N m \overline{v^2}}{V}.$$

rms Speed

In terms of the speed of the gas molecules, the pressure exerted by n moles of an ideal gas is

$$p = \frac{nMv_{\text{rms}}^2}{3V},$$

where $v_{\text{rms}} = \sqrt{\left(v^2\right)_{\text{avg}}}$ is the root-mean-square speed of the molecules, M is the molar mass, and V is the volume.

The rms speed can be written in terms of the temperature as

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}. \qquad v_{\text{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m}}.$$

Table 19-1 Some RMS Speeds at Room Temperature (T = 300 K)[a]

Gas	Molar Mass $\left(10^{-3}\,\mathrm{kg/mol}\right)$	v_{rms} (m/s)
Hydrogen (H_2)	2.02	1920
Helium (He)	4.0	1370
Water vapor (H_2O)	18.0	645
Nitrogen (N_2)	28.0	517
Oxygen (O_2)	32.0	483
Carbon dioxide (CO_2)	44.0	412
Sulfur dioxide (SO_2)	64.1	342

[a]For convenience, we often set room temperature equal to 300 K even though (at 27°C or 81°F) that represents a fairly warm room.

Example: Average speed and rms speed.

Eight particles have the following speeds, given in m/s: 1.0, 6.0, 4.0, 2.0, 6.0, 3.0, 2.0, 5.0. Calculate (a) the average speed and (b) the rms speed.

Translational Kinetic Energy

The **average translational kinetic energy** is related to the temperature of the gas:

$$v_{\mathrm{rms}} = \sqrt{\overline{v^2}} = \sqrt{\frac{3kT}{m}}.$$

$$K_{\mathrm{avg}} = \frac{3}{2}kT.$$

At a given temperature T, all ideal gas molecules—no matter what their mass—have the same average *translational* kinetic energy—namely, $\frac{3}{2}kT$. When we measure the temperature of a gas, we are also measuring the average translational kinetic energy of its molecules.

Checkpoint 2

A gas mixture consists of molecules of types 1, 2, and 3, with molecular masses $m_1 > m_2 > m_3$. Rank the three types according to (a) average kinetic energy and (b) rms speed, greatest first.

Answer:

(a) 1 = 2 = 3

(b) 3, 2 and 1

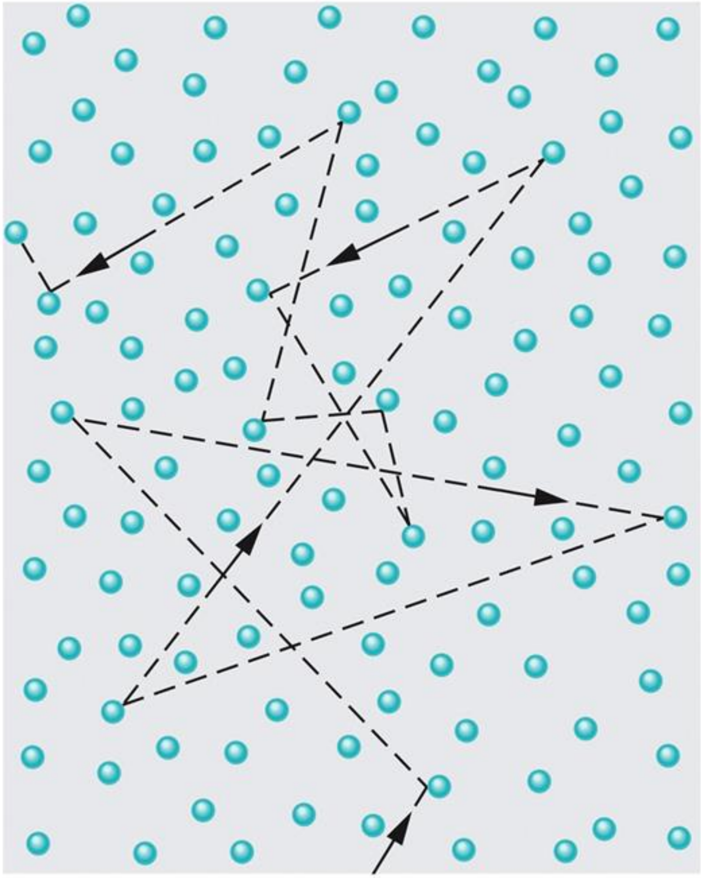
Mean Free Path

The mean free path λ of a gas

is the average distance traversed by a molecule between collisions,

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 \frac{N}{V}}$$

where $\frac{N}{V}$ is the number of molecules per unit volume and d is the molecular diameter.

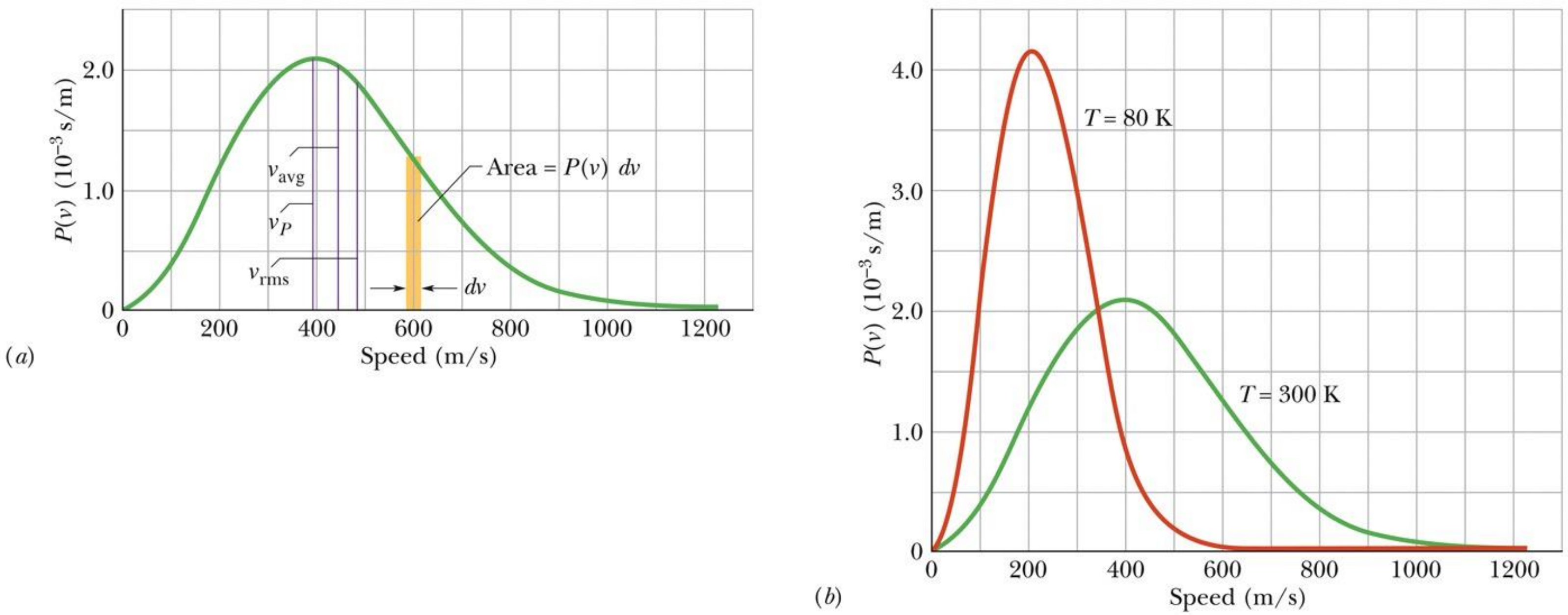


In the figure a molecule traveling through a gas, colliding with other gas molecules in its path. Although the other molecules are shown as stationary, *they are also moving in a similar fashion*.

The Distribution of Molecular Speed

The **Maxwell speed distribution** $P(v)$ is a function such that $P(v)dv$ gives the fraction of molecules with speeds in the interval dv at speed v:

$$P(v) = 4\pi\left(\frac{M}{2\pi RT}\right)^{\frac{3}{2}} v^2 e^{\frac{-Mv^2}{2RT}}.$$



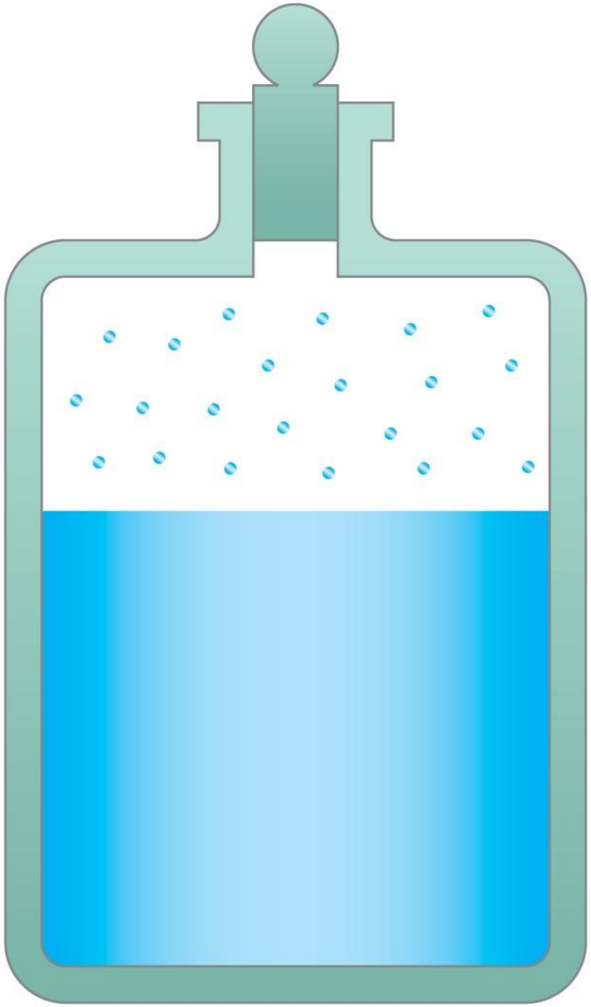
Three measures of the distribution of speeds among the molecules of a gas:

$$v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}} \quad (\text{average speed}),$$

$$v_P = \sqrt{\frac{2RT}{M}} \quad (\text{most probable speed}),$$

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \quad (\text{rms speed}).$$

Vapor Pressure and Humidity



An open container of water can **evaporate**, rather than boil, away. The **fastest** molecules are escaping from the water's surface, so evaporation is a **cooling** process as well.

The inverse process is called **condensation**.

When the evaporation and condensation processes are in **equilibrium**, the vapor just above the liquid is said to be **saturated**, and its pressure is the **saturated vapor pressure**.

A liquid **boils** when its saturated vapor pressure equals the **external** pressure.

Partial pressure is the pressure each component of a mixture of gases would exert if it were the only gas present. The partial pressure of water in the air can be as low as zero, and as high as the saturated vapor pressure at that temperature.

Relative humidity is a measure of the saturation of the air.

$$\text{Relative humidity} = \frac{\text{partial pressure of } H_2O}{\text{saturated vapor pressure of } H_2O} \times 100\%.$$

When the humidity is high, it feels muggy; it is hard for any more water to evaporate.

The dew point is the temperature at which the air would be saturated with water.

If the temperature goes below the dew point, dew, fog, or even rain may occur.

Conceptual Example: Dryness in winter.

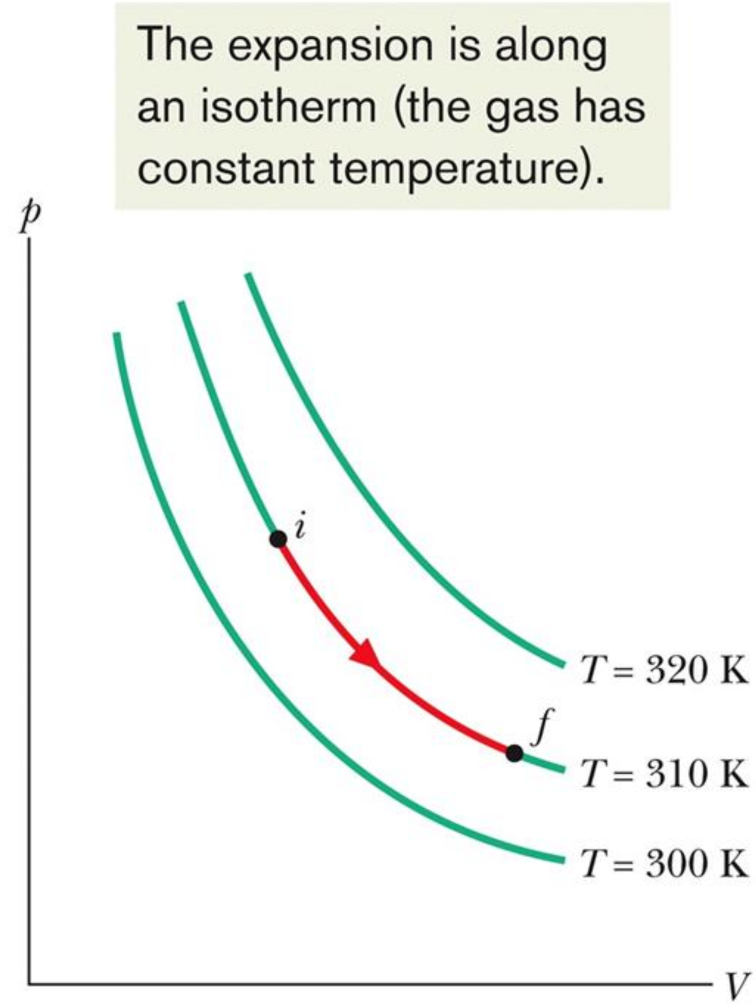
Why does the air inside heated buildings seem very dry on a cold winter day?

Ideal Gases and Work

The work done by an ideal gas during an isothermal (constant-temperature) change from volume V_i to volume V_f is

$$W = nRT \ln \frac{V_f}{V_i}$$

Three isotherms on a p-V diagram. The path shown along the middle isotherm represents an isothermal expansion of a gas from an initial state i to a final state f. The path from f to i along the isotherm would represent the reverse process — that is, an isothermal compression.



The Molar Specific Heats of an Ideal Gas

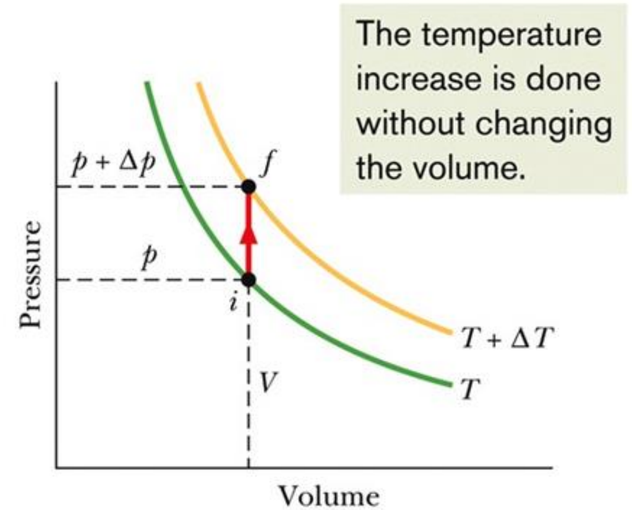
Molar Specific Heat at Constant Volume

The internal energy E_{int} of an ideal gas (monatomic) is a function of the gas temperature only; it does not depend on any other variable.

$$E_{int} = \frac{3}{2}NkT = \frac{3}{2}nRT$$

The **molar specific heat** of a gas at **constant volume** is defined to be

$$Q = nC_V \Delta T$$

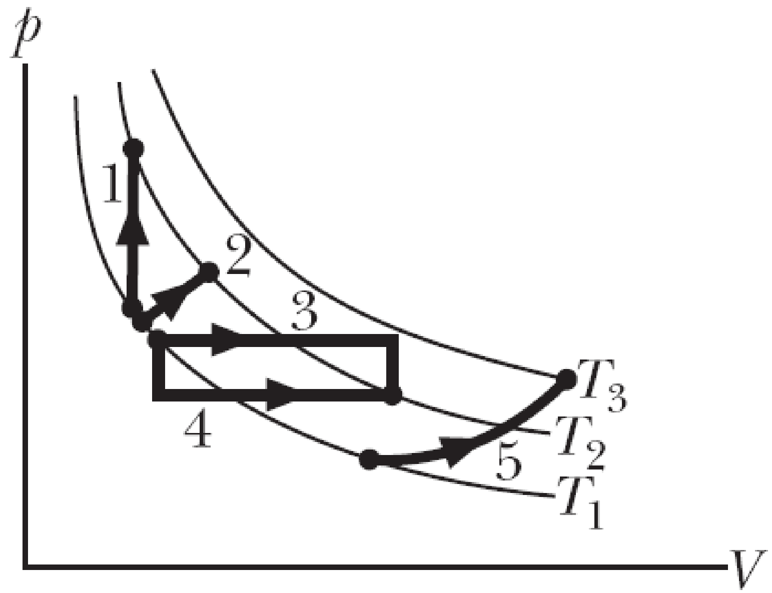


A change in the internal energy E_{int} of a *confined* ideal gas depends on only the change in the temperature, not on what type of process produces the change.

$$\Delta E_{int} = nC_V \Delta T$$

Checkpoint 4

The figure here shows five paths traversed by a gas on a *p*-*V* diagram. Rank the paths according to the change in internal energy of the gas, greatest first.



Answer: 5 and then—4,3,2,1—with same int. energy

Degree of Freedom and Molar Specific Heats

Equipartition of Energy

Table 19-3 Degrees of Freedom for Various Molecules

		Degrees of Freedom			Predicted Molar Specific Heats	
Molecule	**Example**	**Translational**	**Rotational**	**Total (f)**	**C_V (Equation 19-51)**	**$C_p = C_V + R$**
Monatomic	He	3	0	3	$\frac{3}{2}R$	$\frac{5}{2}R$
Diatomic	O_2	3	2	5	$\frac{5}{2}R$	$\frac{7}{2}R$
Polyatomic	CH_4	3	3	6	$3R$	$4R$

Equipartition of energy:

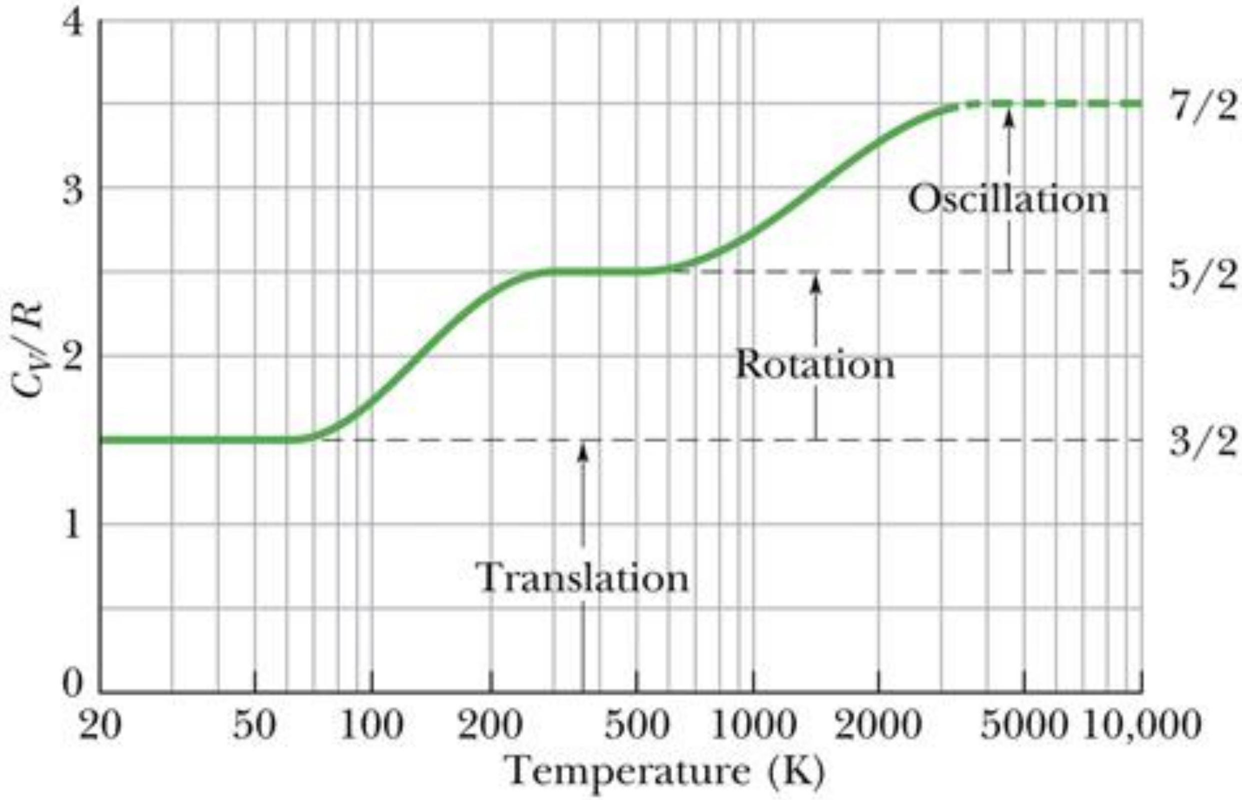
Every kind of molecule has *a certain number (f) of degrees of freedom, which are independent ways in which the molecule can store energy*. Each such degree of freedom has associated with it—on average—an energy of per $\frac{1}{2}kT$ molecule (or $\frac{1}{2}RT$ per mole).

$$C_V = \left(\frac{f}{2}\right)R = 4.16f \text{ J/mol} \cdot \text{K}$$

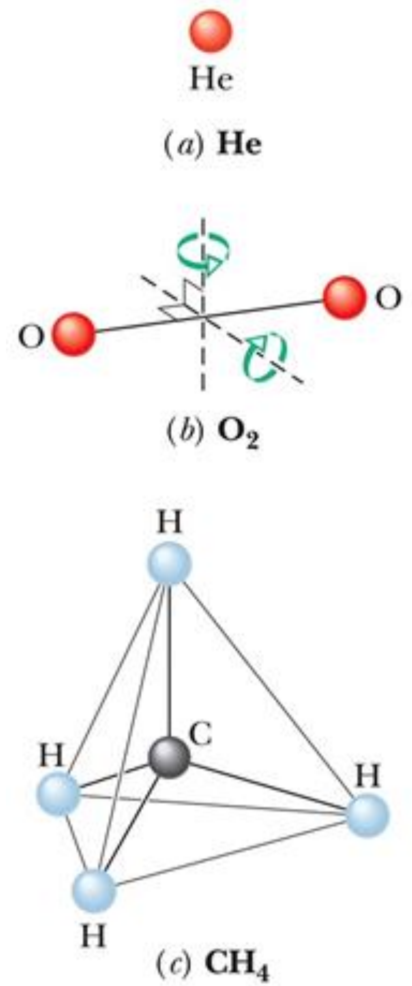
Because rotational and oscillatory motions begin at certain energies, only translation is possible at very low temperatures. As the temperature increases, rotational motion can begin. At still higher temperatures, oscillatory motion can begin.

Gas	C_V/R at 25 °C and 1 atm
Ar	1.50
He	1.50
Ne	1.50
CO	2.50
H_2	2.47
N_2	2.50
O_2	2.53
F_2	2.8
CO_2	3.48
H_2S	3.13
N_2O	3.66

Table 2.3 C_V/R for Various Monatomic, Diatomic, and Triatomic Gases



$\frac{C_V}{R}$ versus temperature for (diatomic) hydrogen gas.



Models of molecule as used in kinetic theory

Learning Outcomes

- ✓ Apply the ideal gas law to solve problems
- ✓ Explain the microscopic view of atom/molecule movement.
- ✓ Explain vapor pressure and humidity
- ✓ Describe Maxwell's distribution and find the rms speed and mean free path.
- ✓ Describe phase transition processes and mole specific heat.
- ✓ Explain the adiabatic process and the Van der Waals equation of state.

$$pV = nRT \quad (\text{ideal gas law}).$$

$$K_{\text{avg}} = \tfrac{3}{2} KT.$$

$$\lambda = \frac{1}{\sqrt{2}\pi d^2 \frac{N}{V}}$$

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}} \quad (\text{rms speed}).$$

$$C_V = \frac{Q}{n\Delta T} = \frac{\Delta E_{\text{int}}}{n\Delta T}$$

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CHAPTER REVIEW AND EXAMPLES, DEMOS

https://openstax.org/books/university-physics-volume-2/pages/2-summary

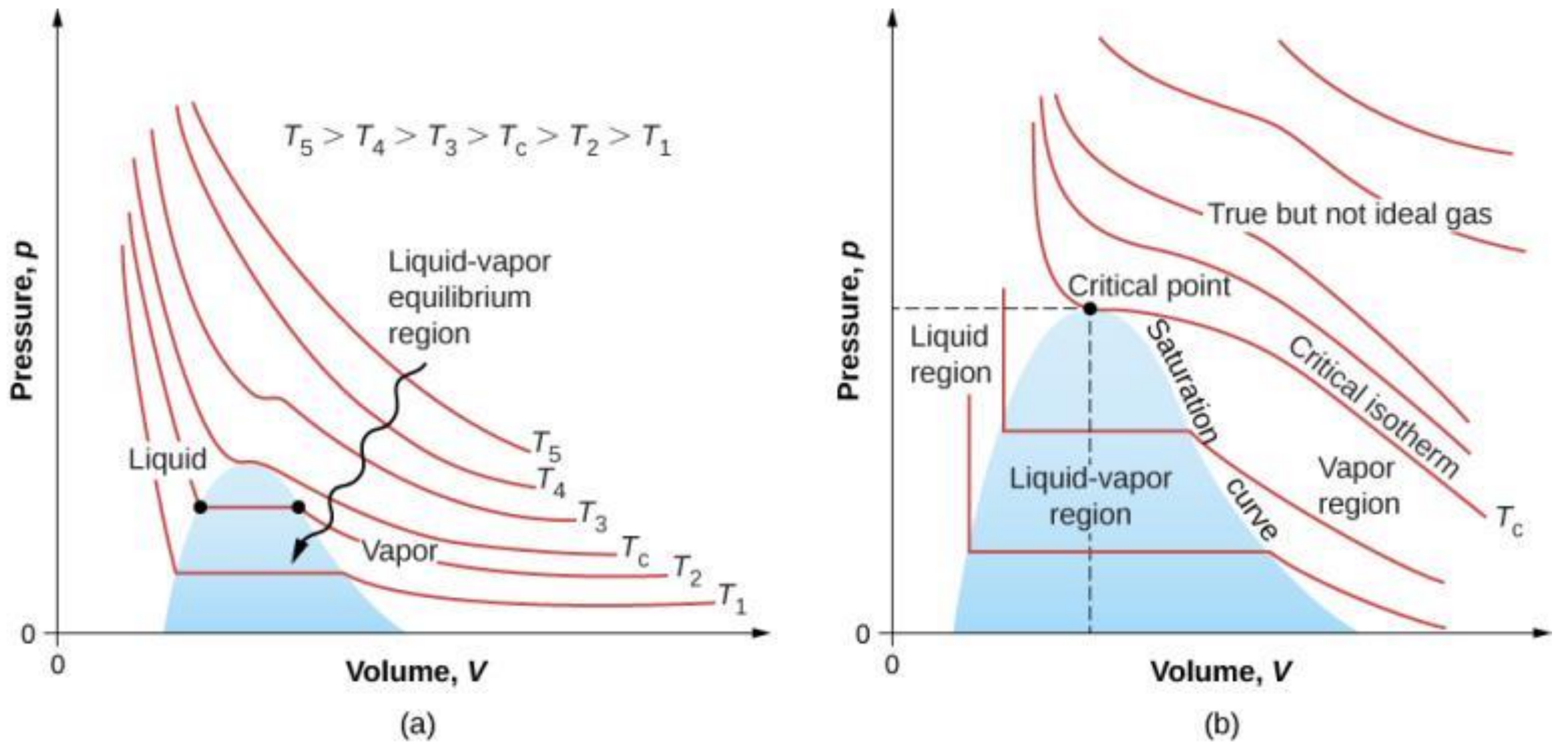
It is known experimentally that for gases at low density (such that their molecules occupy a negligible fraction of the total volume) and at temperatures well above the boiling point, these proportionalities hold to a good approximation. … **IDEAL GAS LAW**

… at *standard temperature and pressure* (STP), which is defined as 0ºC and atmospheric pressure.

The unit of that kind accepted for use with the SI is the *unified atomic mass unit* (u), also called the *dalton*. $6.02\times10^{23}u=1g$.

The ideal gas law is closely related to energy: The units on both sides of the equation are joules. The right-hand side of the ideal gas law equation is Nk_BT. This term is roughly the total translational kinetic energy (which, when discussing gases, refers to the energy of translation of a molecule, not that of vibration of its atoms or rotation) of N molecules at an absolute temperature T, as we will see formally in the next section. The left-hand side of the ideal gas law equation is pV. As mentioned in the example on the number of molecules in an ideal gas, pressure multiplied by volume has units of energy. The energy of a gas can be changed when the gas does work as it increases in volume, something we explored in the preceding chapter, and the amount of work is related to the pressure. This is the process that occurs in gasoline or steam engines and turbines, as we’ll see in the next chapter.

FIGURE 2.8



pV diagrams.

(a) Each curve (isotherm) represents the relationship between *p* and *V* at a fixed temperature; the upper curves are at higher temperatures. The lower curves are not hyperbolas because the gas is no longer an ideal gas.

(b) An expanded portion of the *pV* diagram for low temperatures, where the phase can change from a gas to a liquid. The term "vapor" refers to the gas phase when it exists at a temperature below the boiling temperature.

We digress for a moment to answer a question that may have occurred to you: When we apply the model to atoms instead of theoretical point particles, does rotational kinetic energy change our results? To answer this question, we have to appeal to quantum mechanics. In quantum mechanics, rotational kinetic energy cannot take on just any value; it's limited to a discrete set of values, and the smallest value is inversely proportional to the rotational inertia. The rotational inertia of an atom is tiny because almost all of its mass is in the nucleus, which typically has a radius less than 10^{-14}m. Thus the minimum rotational energy of an atom is much more than $(1/2)k_BT$ for any attainable temperature, and the energy available is not enough to make an atom rotate. We will return to this point when discussing diatomic and polyatomic gases in the next section.

Example 2.5

Calculating Temperature: Escape Velocity of Helium Atoms

To escape Earth's gravity, an object near the top of the atmosphere (at an altitude of 100 km) must travel away from Earth at 11.1 km/s. This speed is called the *escape velocity*. At what temperature would helium atoms have an rms speed equal to the escape velocity?

The mean free path is the length λ such that the expected number of other molecules in a cylinder of length λ and cross-section $4\pi r^2$ is 1. If we temporarily ignore the motion of the molecules other than the one we're looking at, the expected number is the number density of molecules, N/V, times the volume, and the volume is $4\pi r^2 \lambda$, so we have $(N/V)4\pi r^2 \lambda = 1$, or

$$\lambda = \frac{V}{4\pi r^2 N}.$$

Taking the motion of all the molecules into account makes the calculation much harder, but the only change is a factor of $\sqrt{2}$. The result is

$$\lambda = \frac{V}{4\sqrt{2}\pi r^2 N}. \quad (2.10)$$

In an ideal gas, we can substitute $V/N = k_B T/p$ to obtain

$$\lambda = \frac{k_B T}{4\sqrt{2}\pi r^2 p}. \quad (2.11)$$

The **mean free time** τ is simply the mean free path divided by a typical speed, and the usual choice is the rms speed. Then

$$\tau = \frac{k_B T}{4\sqrt{2}\pi r^2 p v_{\text{rms}}}. \quad (2.12)$$

Example 2.9

Calculating Temperature: Calorimetry with an Ideal Gas

A 300-g piece of solid gallium (a metal used in semiconductor devices) at its melting point of only $30.0\ °C$ is in contact with 12.0 moles of air (assumed diatomic) at $95.0\ °C$ in an insulated container. When the air reaches equilibrium with the gallium, 202 g of the gallium have melted. Based on those data, what is the heat of fusion of gallium? Assume the volume of the air does not change and there are no other heat transfers.

Example 2.10

Calculating the Ratio of Numbers of Molecules Near Given Speeds

In a sample of nitrogen $(N_2,$ with a molar mass of 28.0 g/mol) at a temperature of $273\ °C$, find the ratio of the number of molecules with a speed very close to 300 m/s to the number with a speed very close to 100 m/s.

Questions